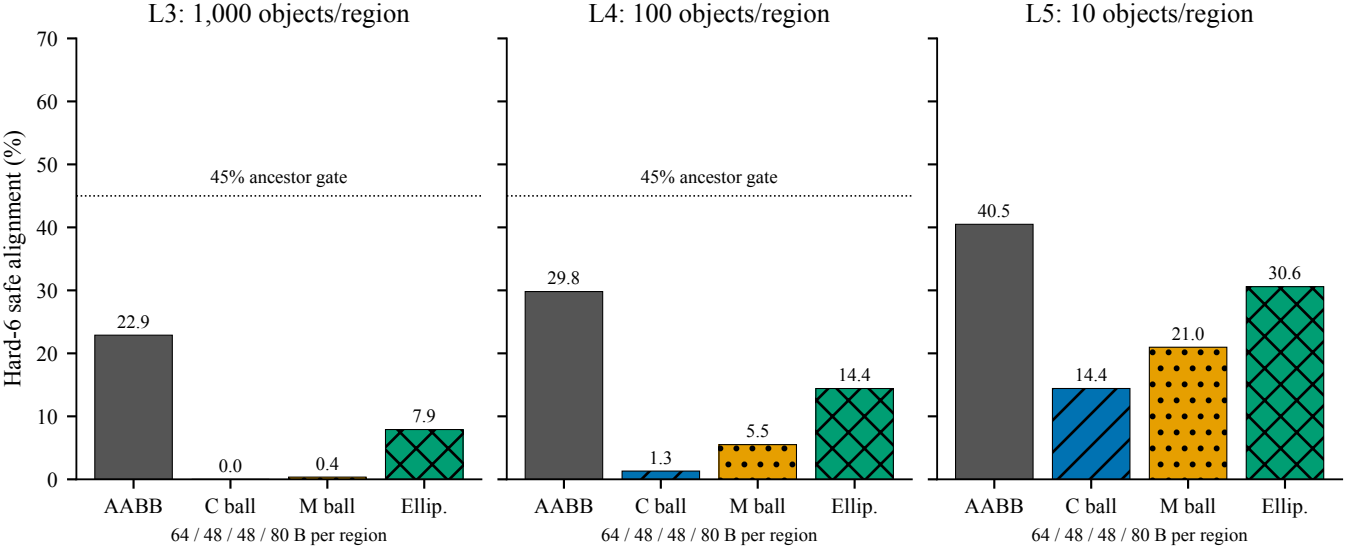
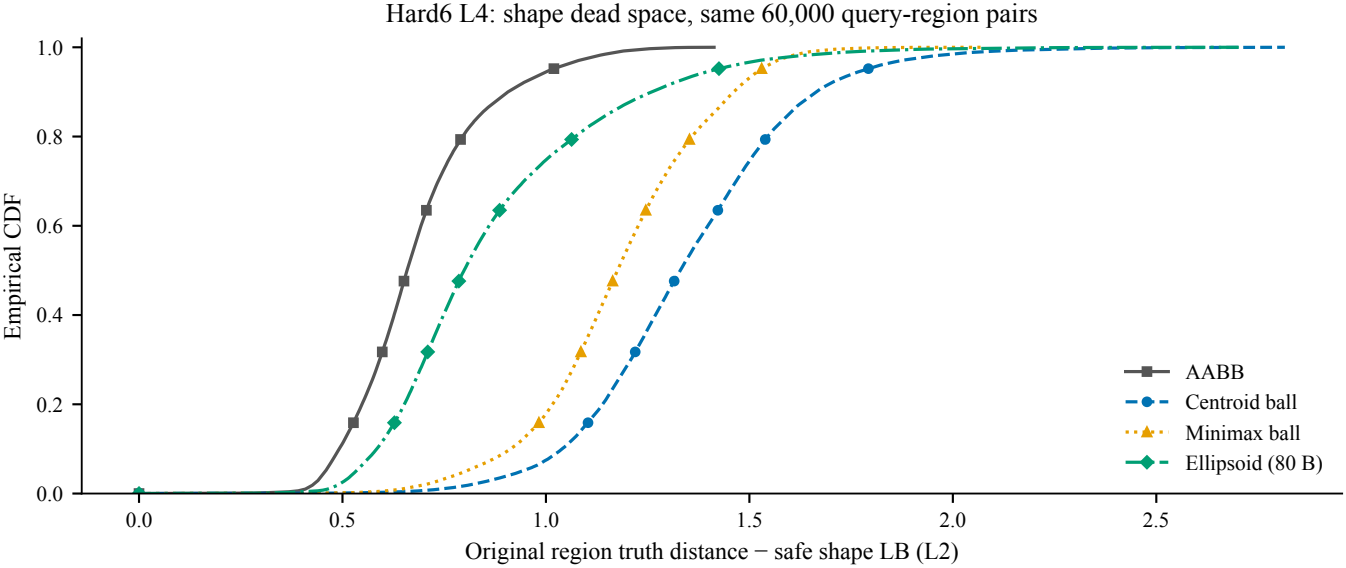
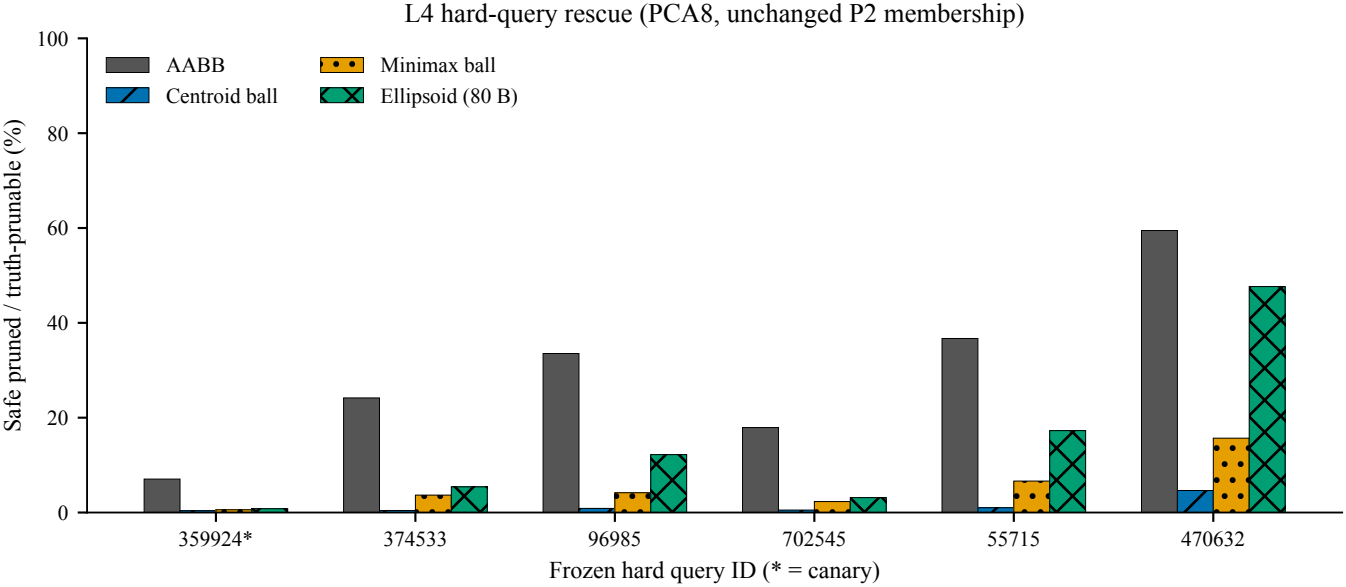
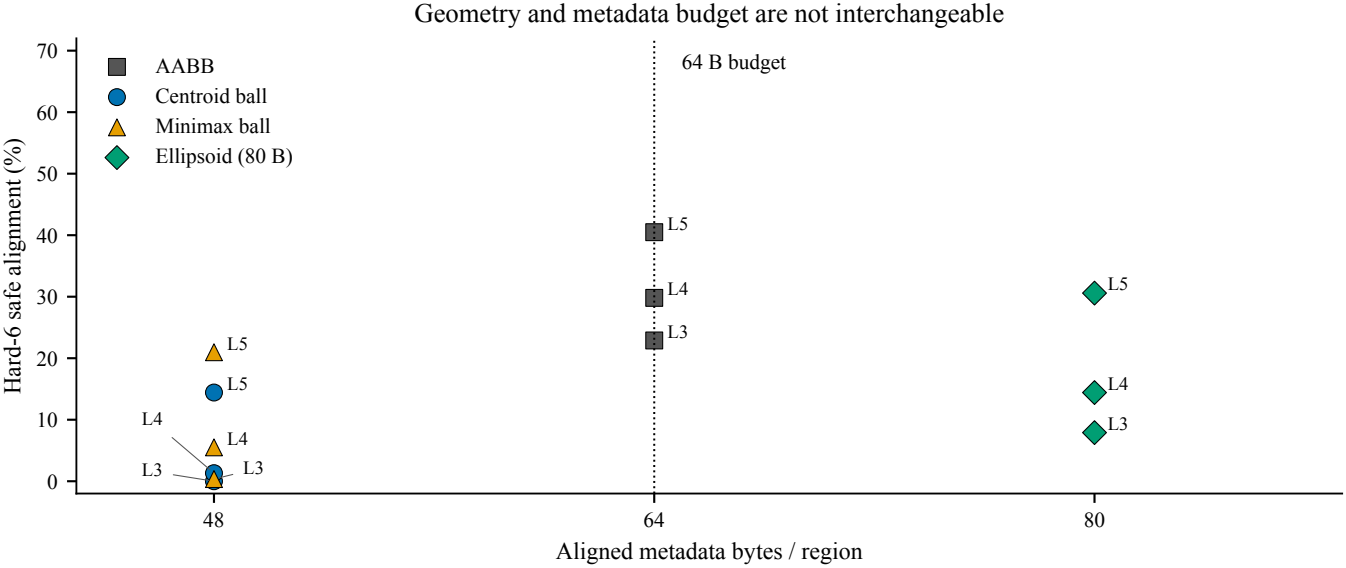

L3: 1,000 objects/region
L4: 100 objects/region
L5: 10 objects/region
Hard-6 safe alignment (%)
70
60
50
40
30
20
10
0
45% ancestor gate
45% ancestor gate
22.9
0.0
0.4
7.9
29.8
1.3
5.5
14.4
40.5
14.4
21.0
30.6
AABB
C ball
M ball
Ellip.
64 / 48 / 48 / 80 B per region


Hard6 L4: shape dead space, same 60,000 query-region pairs
Empirical CDF
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.5
1.0
1.5
2.0
2.5
Original region truth distance − safe shape LB (L2)
AABB
Centroid ball
Minimax ball
Ellipsoid (80 B)


L4 hard-query rescue (PCA8, unchanged P2 membership)
AABB
Centroid ball
Minimax ball
Ellipsoid (80 B)
Safe pruned / truth-prunable (%)
100
80
60
40
20
0
359924*
374533
96985
702545
55715
470632
Frozen hard query ID (* = canary)


Geometry and metadata budget are not interchangeable
AABB
Centroid ball
Minimax ball
Ellipsoid (80 B)
64 B budget
Hard-6 safe alignment (%)
Aligned metadata bytes / region
0
10
20
30
40
50
60
70
48
64
80
L5
L5
L4
L4
L3
L3
L5
L4
L3
L5
L4
L3